

WireSense : An optimization of Aluminium wire rod production in NALCO industry



Department of Computer Science & engineering
IET Lucknow

Submitted By:

Name 1 : Shivansh Singhal (2200520100058)

Name 2 : Muskan Jain (2200520100039)

Name 3 : Divya Prakash Pandey (2200520100024)

Guided by:

Name of supervisor :

Dr. Manik Chandra

Name of co- supervisor :

Mr. Hanuman Maurya

Ms. Kajal

Agenda

- Introduction
- Objective & Problem Statement
 - Literature Survey
- Methodology
- Implementation
- Result
- Conclusion
- Future Scope
- References



Introduction



Machine Learning Relevance

The project uses advanced machine learning models like XGBoost and modified CycleGAN to predict material properties and optimize manufacturing parameters. This enables accurate, fast, and data-driven industrial decision-making.

Interdisciplinary Approach

WireSense combines concepts from Computer Science, Machine Learning, Data Analytics, and Metallurgical Engineering. The project integrates industrial process knowledge with AI-based predictive systems to create a smart manufacturing solution.



Objective & Problem Statement

Problem Statement

Prediction of Aluminium wire rod physical properties through AI, ML or any modern technique for better productivity and quality control.

Study Objectives

WireSense aims to optimize aluminium wire rod production at NALCO using machine learning techniques. It predicts important material properties like UTS, elongation, and conductivity, while also suggesting optimal process parameters to improve product quality, consistency, and industrial efficiency.



FEATURES OF WIRESENSE



PROPERTIES PREDICTOR

Predicts desired properties based on 21 provided parameters.



REVERSE PREDICTOR

The predictor estimates 16 parameters based on 3 properties provided by the user.

Literature Survey

- Prior studies show tree-based ML models (Random Forest, Gradient Boosting, XGBoost) provide high accuracy for predicting metallurgical properties like UTS, elongation, conductivity.
 - Literature highlights XGBoost as the most effective model for nonlinear, tabular industrial datasets due to strong regularization, fast training, and interpretability.
 - Research on industrial ML stresses explainability, using feature importance and SHAP values to help engineers understand how process parameters affect output quality.
 - Studies also address challenges of multi-source industrial data, recommending timestamp alignment, coil/cast ID merging, and anomaly handling, which aligns with WireSense's preprocessing pipeline.
 - Recent works use Optuna-based hyperparameter tuning and evaluate performance with MAE/RMSE, similar to the WireSense methodology.
-

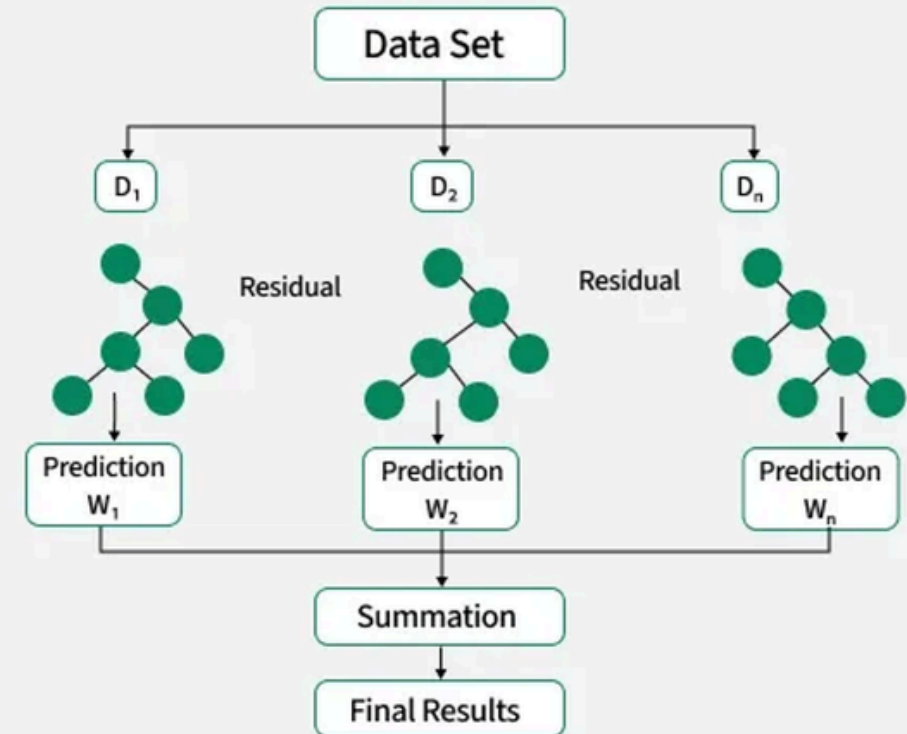
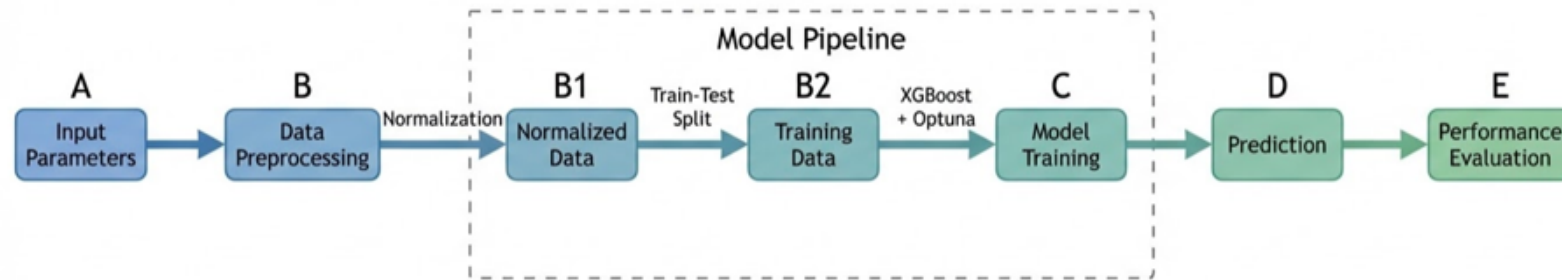
Existing / Proposed Methodology

=

PROPERTIES PREDICTOR

XGBOOST MODEL

XGBoost, or Extreme Gradient Boosting, is a machine learning library that uses gradient boosted decision trees (GBDTs): Combines weak learner trees into strong learners by adding up residuals thus works on principle of ensembling



WHY XGBOOST?

FLEXIBILITY

Handles irregular patterns and uninformative features effectively.

ACCURACY

Achieves better performance and generalization on medium-sized datasets.

SELECTIVE

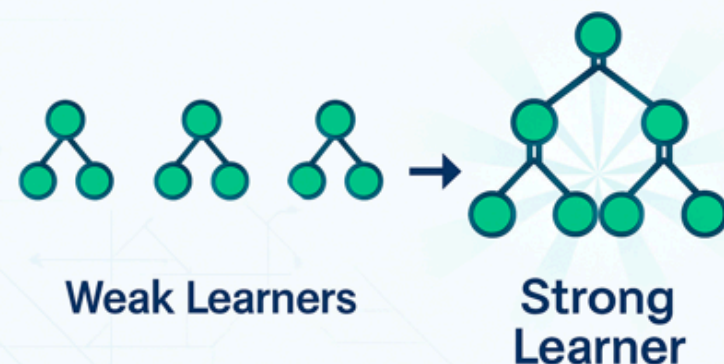
Reduces overfitting by focusing only on informative features.

SCALABLE

Scalable for large datasets with millions of records.



XGBoost

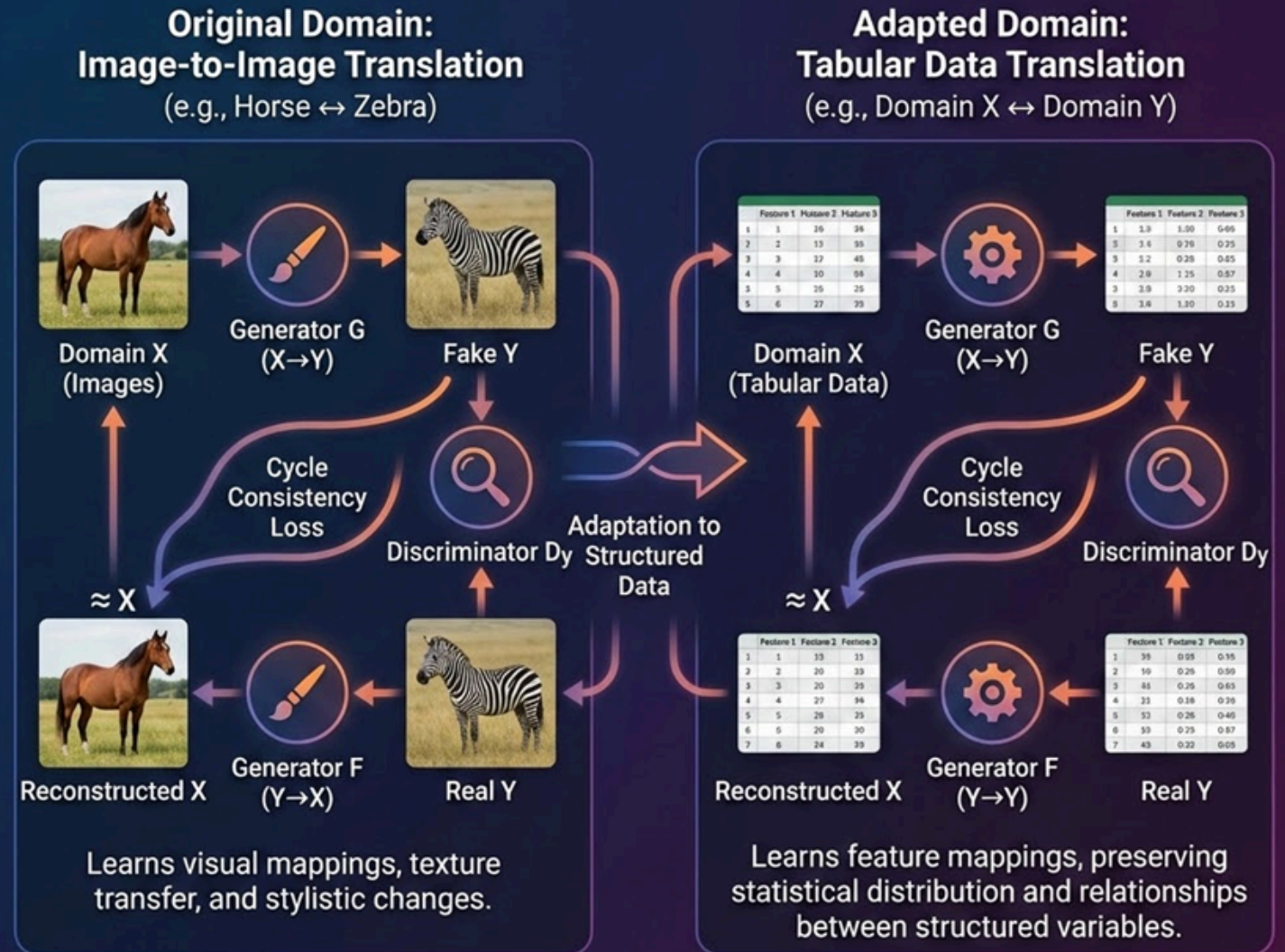


REVERSE PREDICTOR

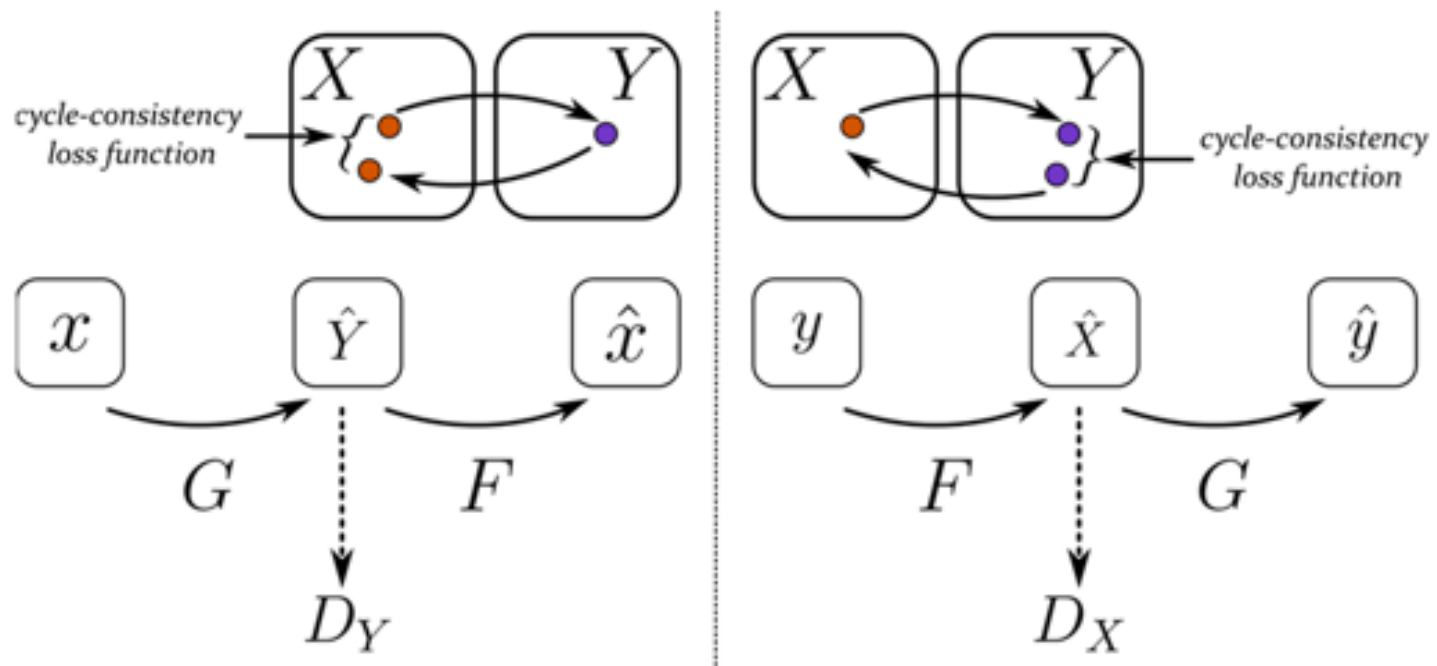
MODIFIED CYCLEGAN

It consist of four main components: two generators and two discriminators. These components work together with adversarial loss and cycle consistency loss to perform from X to Y and Y to X.

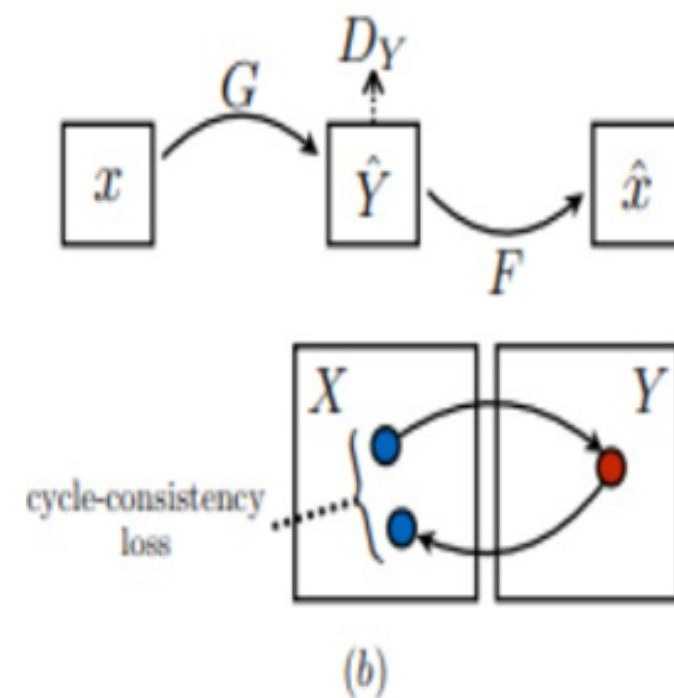
CycleGAN: From Pixels to Structured Data



Key Adaptation: The core CycleGAN architecture (two generators, two discriminators, cycle-consistency loss) remains, but the networks are designed to process and generate structured, high-dimensional feature vectors instead of 2D image grids, focusing on preserving data integrity and statistical properties rather than visual appearance.



MODIFIED CYCLEGAN



Data Processing and Merging Pipeline

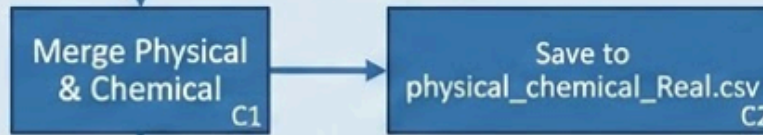
Initial Data Loading



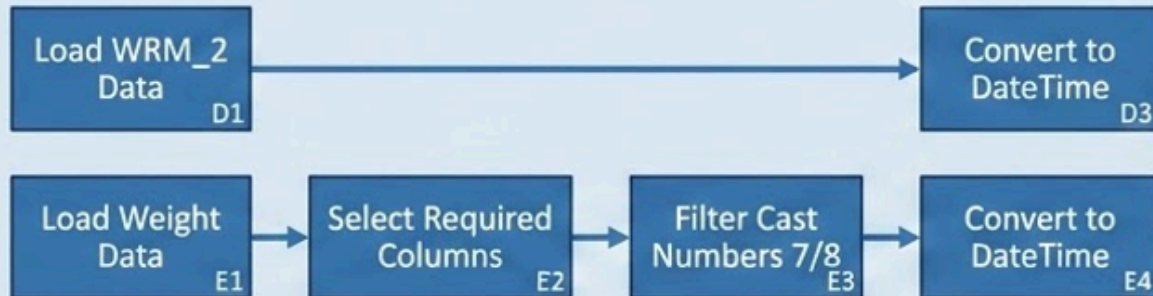
Physical Data Processing



First Merge



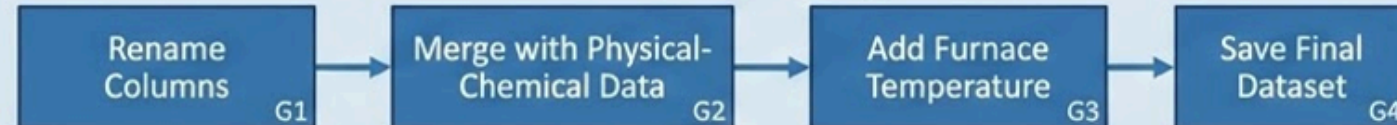
WRM and Weight Data Processing



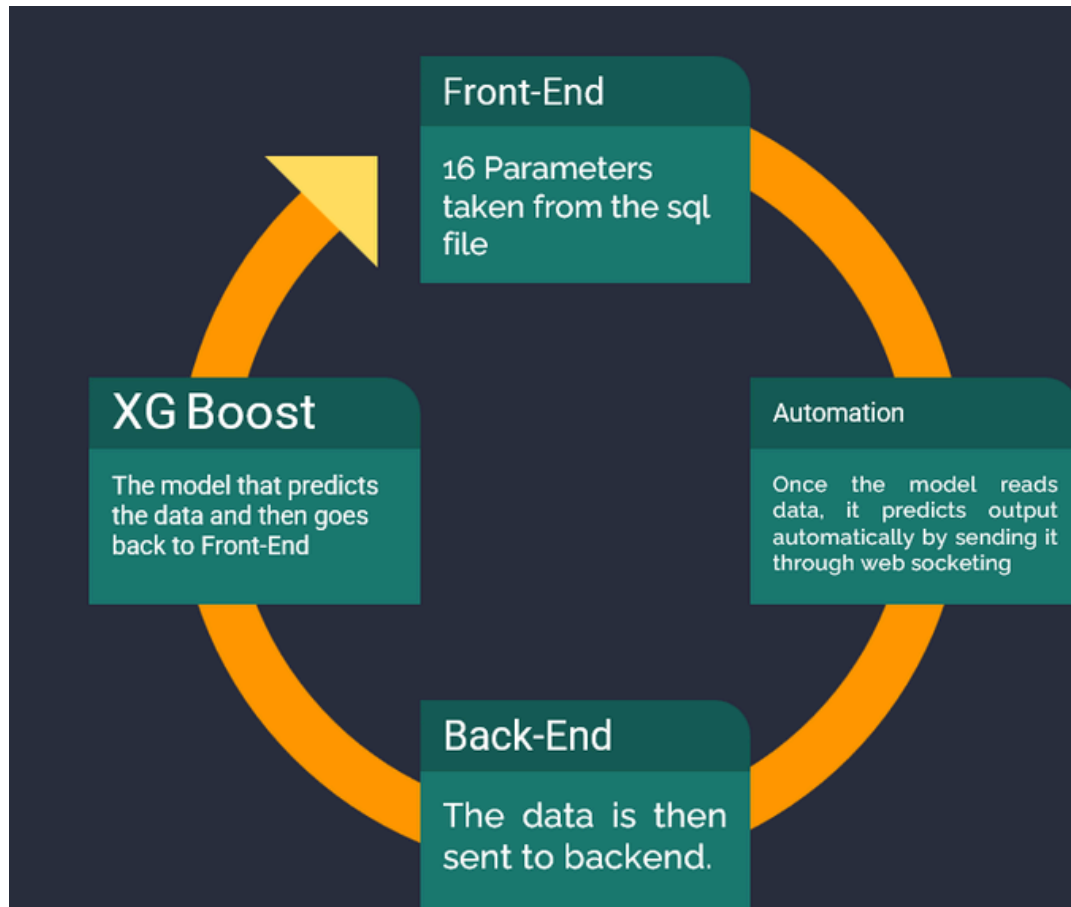
Time-Based Merge



Final Processing



Implementation

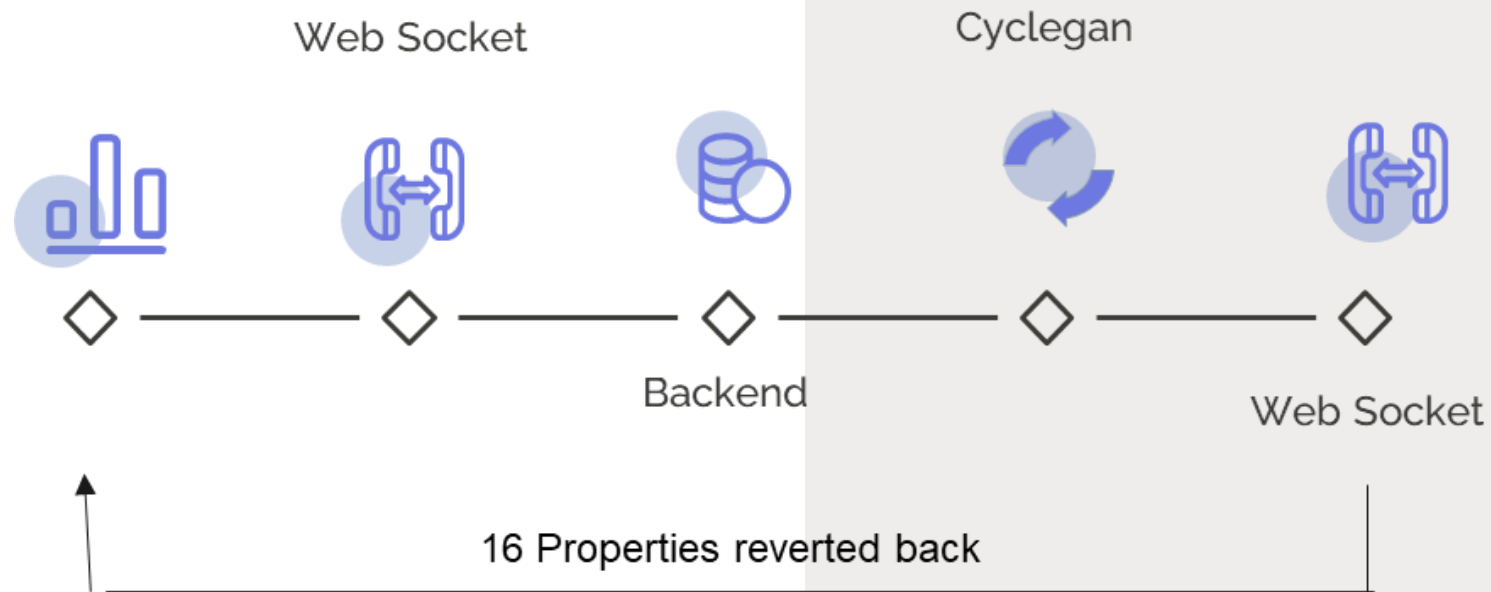


PROPERTIES PREDICTOR

Implementation

=

REVERSE PREDICTOR



Result



MSE

UTS=0.00135

elongation=0.00840

CONDUCTIVITY=0.005106



ERROR PERCENTAGE

UTS=3.64%

ELONGATION=3.41%

CONDUCTIVITY=0.17%



MAD

UTS= 0.3429723519545335

elongation=1.5768071143181772

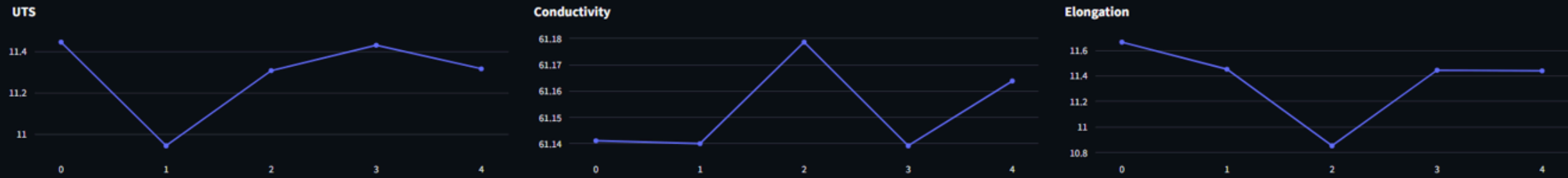
CONDUCTIVITY= 0.10346717834472634

Parameters Live Monitor

FINAL
WORKING
DASHBOARD



Predictions on Current Window



Intial Parameter Predictor

Move the sliders to set your target UTS, Elongation, and Conductivity. Click Predict and the Cycle-GAN generator will output the 16 upstream process parameters likely to achieve those targets.



 Predict

	Predicted Value
EMUL_OIL_I_TEMP_PV_VAL0	60.3102
STAND_OIL_I_TEMP_PV_REAL_VAL0	54.3038
GEAR_OIL_I_TEMP_PV_REAL_VAL0	62.3178
EMUL_OIL_I_PR_VAL0	2.2441
QUENCH_CW_FLOW_EXIT_VAL0	9.2187
CAST_WHEEL_RPM_VAL0	2.8813
BAR_TEMP_VAL0	581.2614
QUENCH_CW_FLOW_ENTRY_VAL0	242.2461
GEAR_OIL_I_PR_VAL0	1.8653
STANDS_OIL_I_PR_VAL0	1.9013
TUNDISH_TEMP_VAL0	713.9071
RM_MOTOR_COOL_WATER__VAL0	1.1644
ROLL_MILL_AMPS_VAL0	235.6201
RM_COOL_WATER_FLOW_VAL0	208.7776
EMULSION_LEVEL_ANAL0_VAL0	1003.7106
Furnace_Temperature	576.6272

FINAL WORKING DASHBOARD

Dashboard

Activity

Library

Security

Schedules

Payouts

Settings

CONDUCTIVITY
65 s/m



UTS
35 n/m²

ELONGATION
20 %



65 s/m

Conductivity ▲ +2.45%



Alerts ☐

- Conductivity**
Increase Temperature by 2 % +0.5
- UTS**
Decrease Pressure by 3 atm -1.4
- Elongation**
No changes required +0.01

[View all →](#)

Casting Temperature

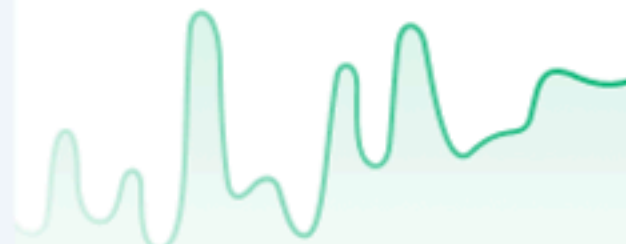
435 K

▲ +2.45%



Casting Pressure

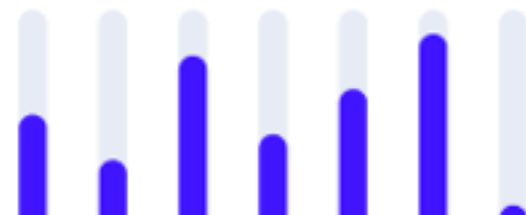
▲ +7.45%



Rolling Speed

✓ On track

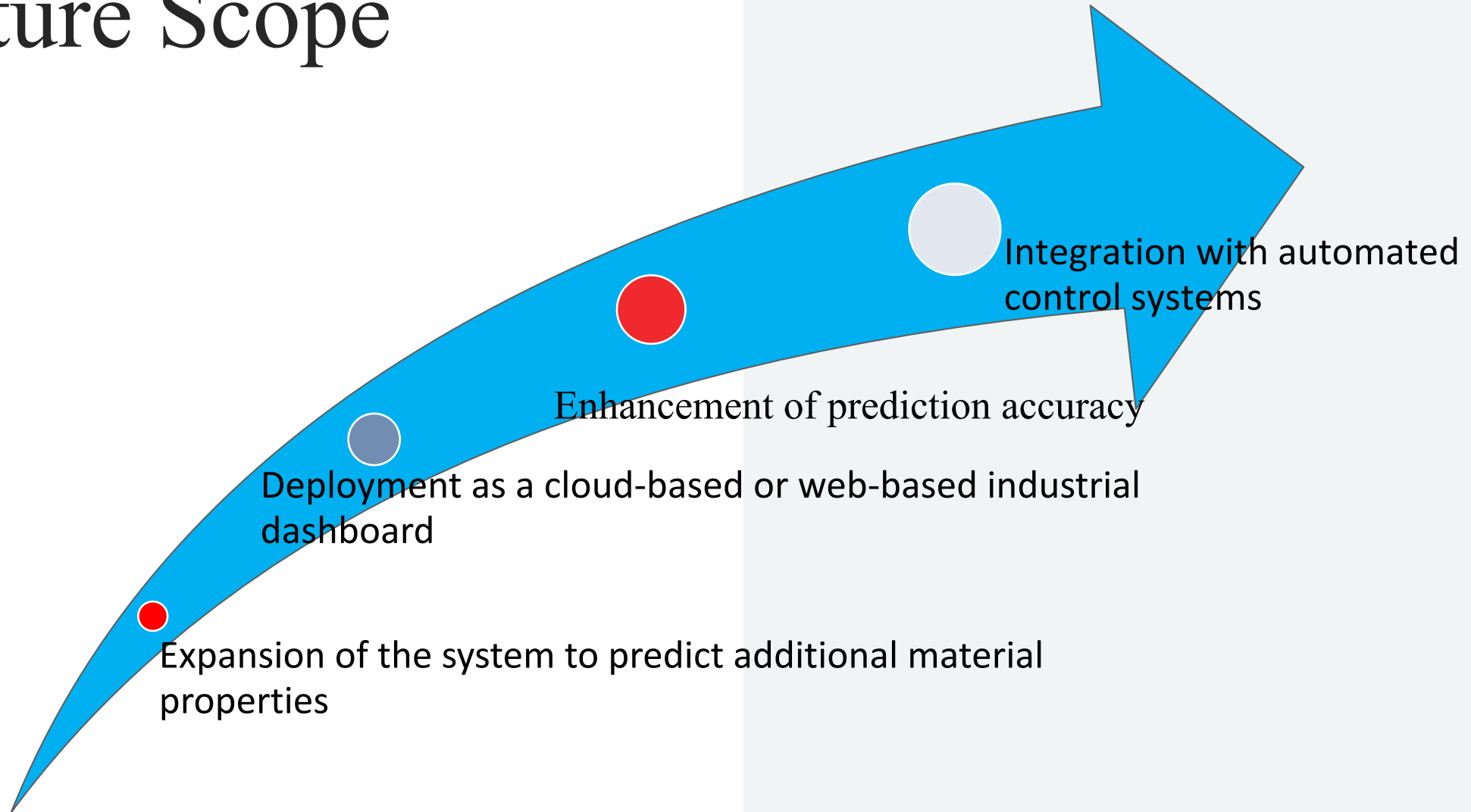
▲ +2.45%



Conclusion

- WireSense provides a data-driven solution to improve aluminium wire rod manufacturing at NALCO using machine learning techniques.
 - The XGBoost-based predictor accurately estimates key material properties such as UTS, elongation, and conductivity from industrial process parameters.
 - The modified CycleGAN reverse predictor helps determine optimal process parameters from desired output properties.
 - The system reduces manual trial-and-error, improves product consistency, and supports faster industrial decision-making.
 - Overall, WireSense promotes Industry 4.0 practices by enhancing efficiency, reliability, and quality control in metallurgical manufacturing
-

Future Scope



References

- [1] T. Chen and C. Guestrin, “XGBoost: A Scalable Tree Boosting System,” in Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD), 2016. [Online]. Available: <https://arxiv.org/abs/1603.02754>
- [2] J. Y. Zhu, T. Park, P. Isola, and A. A. Efros, “Unpaired Image-to-Image Translation using Cycle-Consistent Adversarial Networks,” in Proceedings of the IEEE International Conference on Computer Vision (ICCV), 2017. [Online]. Available: <https://arxiv.org/abs/1703.10593>
- [[3] T. Akiba, S. Sano, T. Yanase, T. Ohta, and M. Koyama, “Optuna: A Next-generation Hyperparameter Optimization Framework,” in Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD), 2019. [Online]. Available: <https://arxiv.org/abs/1907.10902>
- [4] Z. Lu et al., “Machine learning driven design of high-performance Al alloys,” Journal of Materials Informatics, 2024. <https://www.oaepublish.com/articles/jmi.2024.21> [Online]. Available:
- [5] S. M. Lundberg and S. I. Lee, “A Unified Approach to Interpreting Model Predictions,” in Advances in Neural Information Processing Systems (NeurIPS), 2017. [Online]. Available: <https://arxiv.org/abs/1705.07874>

Thank you